

Multidirectional Instability of the Shoulder

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The shoulder is unique in that it is the most mobile joint in the human body. It serves primarily to position the hand in space and thus requires a delicate balance of stability and flexibility to accomplish its function. The inherent complexity of the glenohumeral joint requires contributions from both dynamic and static stabilizers to achieve this balance. Further, whereas some activities, such as swimming, place a premium on flexibility, others, such as lifting a weight overhead, value stability more highly. Shoulder instability is a spectrum of pathologic conditions in which the humeral head exceeds the functional or physiologic limit of the confines of the glenoid labral complex, resulting in a sensation that is troubling to the patient.

Instability of the shoulder can be in any cardinal direction or in a combination of directions. In 1980 Neer and Foster¹ introduced the term *multidirectional instability* (MDI), which refers to anterior and posterior instability associated with involuntary inferior subluxation or dislocation. Matsen² subsequently categorized shoulder instability into two general types: (1) traumatic and unidirectional with a Bankart lesion treated surgically, and (2) atraumatic, multidirectional, and bilateral, responds to rehabilitation, and may require an inferior capsular shift for surgical treatment. Although these categories are helpful for general classification, one must be careful not to oversimplify MDI. Patients with MDI can certainly present with unilateral symptoms, and although it is less common, they may present only after a traumatic event. As a result, both Neer's description of MDI and Matsen's description of the second general type have been modified to reflect our increasing understanding of the multidirectionally unstable shoulder.

It is important to consider the contributions of both normal anatomy and functional demands to more fully understand MDI.³ Laxity is highly variable within the shoulder and thus, in and of itself, is not equal to instability. Many patients use an increase in laxity to optimize performance; for example, laxity provides elite swimmers with a competitive advantage. These athletes rely on a dynamic stability provided by the shoulder and periscapular musculature to allow extreme ranges of motion without pathologic instability.⁴ Thus physical signs of laxity, such as the sulcus sign, or the ability of a patient to demonstrate subluxation, should not be equated with shoulder instability.^{5,6} Without a precise definition of MDI, it is critically important to use all the tools available to the treating physician, including history, physical examination, imaging, examination while the patient is anesthetized, and diagnostic arthroscopy, to ensure

that the appropriate anatomic deficiency of patients with MDI is addressed.

Glenohumeral stability results from a complex interplay of static and dynamic stabilizers. Static stabilization of the glenohumeral joint is controlled by such factors as the native depth of the glenoid and surrounding labral complex, as well as the length and stiffness of the shoulder capsule and supporting ligaments. Although the depth of the glenoid has been shown to provide a small restraint to glenohumeral translation, and patients with MDI demonstrate glenoid cavities that are more shallow than those of age-matched control subjects,⁷⁻⁹ the characteristic pathologic entity in persons with MDI is increased capsular redundancy.^{1,10,11} This redundancy may be congenital and associated with a systemic disease, such as Ehlers-Danlos syndrome,¹² osteogenesis imperfecta,¹³ Marfan syndrome,¹⁴ benign joint hypermobility syndrome, and fascioscapulohumeral dystrophy, or it can be acquired from repeated minor injuries (microtrauma) or repetitive use during exercises that stretch the capsuloligamentous restraints. Different parts of the capsule have been shown to be responsible for glenohumeral stability because the position of the shoulder and the direction of opposing forces are varied. The anterior and posterior bands of the inferior glenohumeral ligament act as a hammock to balance the humeral head on the glenoid.¹⁵⁻¹⁷ The most significant restraint to inferior subluxation of the humeral head is the rotator interval complex, consisting of the superior glenohumeral ligament and coracohumeral ligament together with the inferior glenohumeral ligament complex.^{18,19}

The shoulder is also dynamically stabilized by the rotator cuff, deltoid, and long head of the biceps, which collectively serve to compress the humeral head into the glenoid in a phenomenon coined *concavity compression*. One study demonstrated that the shoulder could resist translational loads equal to 60% of the compressive load applied.²⁰ In addition, the periscapular musculature can affect the position of the scapula, effectively altering glenoid version and inclination. Although the role of these dynamic stabilizers is more difficult to study, abnormal scapular kinematics and muscle activation patterns have been demonstrated in patients with MDI compared with asymptomatic control subjects.^{9,21-23} One explanation of the etiology of MDI is the combination of baseline hyperlaxity and acquired scapular muscle dysfunction. This theory may explain why some swimmers present with impingement syndrome, why some baseball pitchers experience internal impingement, and

Abstract

Multidirectional instability (MDI) has been defined as involuntary inferior subluxation with instability in at least one other direction, anterior and/or posterior. The root cause of MDI results from a disruption in the balance of static and dynamic shoulder stabilizers. This can be due to various pathologies, but capsular redundancy and a hyperlax shoulder with instability symptoms are the most important aspects of the MDI diagnosis. A combination of history, physical exam, and some diagnostic studies are used to identify MDI, the instability exam being the most critical. Although physical therapy remains the primary initial treatment in MDI, surgical interventions have shown great success. Open inferior capsular shifts and arthroscopic plications have success rates >90% and recurrence rates <10% in some series. Results for return to sports and the competition levels of the patients studied are both variable. Surgical treatment is a delicate balance between providing stability and preventing a decrease in range of motion. Communication between the physician, therapist, and the patient is essential to patient post-operative management especially for athletes returning to sports. Although the most feared complication is neurological injury, transient neuropraxias are sparse in the literature. Recurrent instability is the most common complication and currently there are no great techniques to solve this difficult problem. While developing a universal classification system is important, identifying risk factors for surgical failure and developing specific return to sport criteria are also pressing future research directions.

Keywords

multidirectional instability
shoulder instability
capsular redundancy
open inferior capsular shift
arthroscopic plication

why other patients with hyperlaxity present with pain as their chief complaint.

HISTORY

MDI can be very challenging to diagnose correctly. Patients usually do not report instability but rather vague pain of insidious onset, and thus the chief complaint is often not suggestive of the diagnosis. Further, although patients usually present with hyperlaxity, it is often not significantly different from their asymptomatic other side, and thus discerning between normal laxity versus instability in a particular patient can be confusing. Nevertheless, the history and physical examination are the critical steps in making an accurate diagnosis, and maintaining a high level of clinical suspicion is essential given the variety of presenting complaints.

Most patients with MDI present with a chief complaint of activity-related pain, but they may also report decreased strength or athletic performance. It is important to identify which activities are causative, because MDI is more common in overhead throwing sports, and the offending activity gives insight into both the focus of treatment and the patient's expectations for return. Reports of pain when carrying a heavy object, such as a suitcase, should raise suspicion. Cofield and Irving²⁴ stressed four questions in the workup of MDI: (1) What is the relationship to trauma? (2) Does the patient intentionally cause the instability? (3) What is the degree of instability? (4) What is the direction of instability? Regarding the first question, most patients with MDI have an atraumatic onset, whereas a single inciting event can be responsible for tipping the patient from hyperlaxity to instability. If one loosens the definition of "trauma" to include repetitive use or conditions that compromise the dynamic stabilizers, one can gain a much deeper understanding of the source of the patient's complaints and a rational approach to addressing them. The second question about volitional instability by Cofield and Irving is also critical. Since publication of the classic article by Rowe and colleagues,²⁵ it has become a tenet that patients with willful dislocation should be approached cautiously. However, as pointed out by Mallon and Speer,¹⁰ it should be remembered that not all patients who have the ability to dislocate the shoulder have a psychologically pathologic condition, which is an important differentiator in the treatment algorithm.

The degree of instability is another important component of the history. Patients with MDI may undergo either recurrent subluxation or dislocation, although the former is more common,¹⁰ and an understanding of the degree of instability may help determine the aggressiveness of the treatment program. When patients report that their joint "slips" out in the absence of trauma or during sleep, clinical suspicion should be heightened. Finally, the question of the primary direction of instability in patients with MDI is critical to making a complete and accurate assessment and correctly designing an approach to treatment. Most patients do not have true global instability, and distinguishing between true MDI and unidirectional instability with multidirectional laxity is one of the critical steps guiding treatment.³ Finally, it is critical to understand the diagnosis in the context of the patient. The type of activity the patient expects to resume

and within what time frame, as well as the management of the individual patient's expectations about rehabilitation and return, are essential in guiding successful treatment.

PHYSICAL EXAMINATION

Foster²⁶ has noted that the most useful tool for making the diagnosis of MDI is the physical examination. Because patients with MDI often have vague complaints, it is essential to perform a thorough physical examination. Generalized ligamentous laxity has long been considered a hallmark of MDI and should be evaluated in each patient. Beighton et al.²⁷ reported on five physical signs of joint hyperlaxity (Table 42.1), which add up to a potential of nine points. Knee, elbow, thumb, and metacarpophalangeal hyperextension are all given a point for each side, along with the patient's ability to place his or her hands flat on the floor with straight knees from a standing position. Any score greater than 3 is considered good evidence of generalized ligamentous laxity and should alert the examiner to the presence of this condition. Other classification systems have been reported for collagen disorders, including the Villefranche nosology²⁸ system for Ehlers-Danlos syndrome subtypes and the Ghent criteria defining Marfan syndrome.²⁹ It is important to recognize these connective tissue disorders, because they have been associated with poorer results after surgical stabilization.³⁰ Finally, hyperelasticity and scarring in skin folds, as well as keloid formation or widening of scars, are also suggestive of a collagen disorder.³¹ The presence of generalized ligamentous laxity is not sufficient for the diagnosis of instability because a wide spectrum of asymptomatic laxities exists in all cardinal directions.^{32–34}

Since the original description by Neer and Foster,¹ the sulcus sign has been considered the *sine qua non* of MDI (Fig. 42.1).³⁵ This test is performed by placing axial traction on the arm in its resting position at the patient's side and can be evaluated with rotation, as well as at varying degrees of abduction. This action should produce a visible and palpable sulcus underneath the acromion as the humerus translates inferiorly. Although Harryman et al.³⁶ noted significant variation in what a "normal"

TABLE 42.1 Beighton Scale for Multidirectional Laxity

Characteristic	Score
Passive dorsiflexion of the little finger beyond 90 degrees	1 point for each hand
Passive apposition of the thumb to the ipsilateral forearm	1 point for each hand
Active hyperextension of the elbow beyond 10 degrees	1 point for each hand
Active hyperextension of the knee beyond 10 degrees	1 point for each hand
Forward flexion of the trunk with extended knees so that the palms of the hands rest flat on the floor	1 point

A score greater than 3 out of a possible 9 points is diagnostic of hyperlaxity.



Fig. 42.1 The sulcus sign.

sulcus is, in a healthy control group, the sulcus measured approximately 11 mm upon testing. Humeral head displacement of more than 2 cm from the acromion, as well as the presence of a sulcus sign with 90 degrees abduction, is indicative of considerable inferior capsular laxity.^{1,7,37} It is again important to heed Matsen's caution that to make the diagnosis of MDI, the sulcus sign must reproduce the patient's symptoms.³⁸

The hyperabduction test as described by Gagey and Gagey³⁹ is another test for inferior glenohumeral laxity in which patients are evaluated for comparative glenohumeral abduction. In asymptomatic volunteers, abduction averages greater than 90 degrees, and a positive test is indicated by passive abduction greater than 105 degrees. Although MDI was an exclusion criterion in the study, Balg and Boileau⁴⁰ found shoulder hyperlaxity anteriorly or inferiorly, as defined by excessive external rotation and a positive hyperabduction test, respectively, to be an independent variable for recurrent instability after an arthroscopic Bankart procedure.

In addition to inferior instability, it is important to examine any patient with suspected MDI for anterior and posterior instability. With the common finding of generalized laxity, it is important that these provocative maneuvers result in the reproduction of the patient's symptoms. The apprehension test⁴¹ is a workhorse test for anterior instability that is performed with the patient either seated or supine. With this test, the examiner takes the patient's arm into maximal external rotation with the shoulder at 90 degrees of abduction. Reproduction of the patient's symptoms here is a reliable indicator of anterior instability.⁴² Another common method of assessment for instability is the abducted anterior load-and-shift test. This test is performed with the patient either in the seated, supine, or lateral decubitus position (Fig. 42.2). This latter position is particularly helpful because the scapula can be controlled with one hand while the humerus is translated with the other. The humeral head is loaded axially to ensure it is centered and then translated forward. The test is graded on the basis of how far the humeral head travels in relation to the glenoid (Table 42.2)⁴³: Grade 0, little to no movement of the humeral head; grade 1, the humeral head rides up to the glenoid labrum; grade 2, the humeral head is shifted off the glenoid but spontaneously reduces when pressure is removed; and grade 3, the humeral head is shifted off the glenoid and remains dislocated once the pressure is removed.



Fig. 42.2 The abducted anterior load-and-shift test. This test can be performed in the lateral decubitus position in a patient who is awake or anesthetized. This position allows the examiner to stabilize the scapula with one hand and translate the humerus with the other.

TABLE 42.2 Grading System for Load and Shift for Instability

Grade	Characteristic Glenoid Position
1	Little to no movement of the humeral head
2	The humeral head rides up to the glenoid labrum
3	The humeral head is shifted off the glenoid but spontaneously reduces when pressure is removed
4	The humeral head is shifted off the glenoid and remains dislocated once the pressure is removed



Fig. 42.3 The push-pull test performed with an anesthetized patient. This test can also be performed reliably in a patient who is awake.

For posterior instability, a similar load and shift maneuver can be performed. We have found that a modification of this test, the push-pull test,⁴⁴ is most effective in controlling posterior translation (Fig. 42.3). The push-pull test is performed with the patient supine; the arm is placed at 90 degrees abduction in neutral rotation and 30 degrees horizontal adduction. The examiner grasps around the patient's wrist with one hand (the pull)

and posteriorly loads the humerus (the push), which places a fulcrum on the shoulder and enhances the examiner's ability to control subluxation. The evaluation of instability in all three directions is essential to determine the most effective treatment regimen for the patient.

It must be remembered that MDI is a combination of static and dynamic restraints, and therefore the physical examination of any patient with suspected MDI must include a thorough evaluation of the patient's musculature. This evaluation begins with visualization of the patient's scapular musculature and kinematics. Patients should be evaluated for atrophy, winging, and scapular motion. Deltoid and rotator cuff strength should be assessed and compared with the opposite side, with particular attention directed to scapular rhythm as the arm moves, specifically through the motion that causes the patient's symptoms. Symptoms can be far better explained by asymmetries in the musculature than by shoulder laxity.

IMAGING

Although MDI is primarily diagnosed clinically, imaging can enhance the work-up in many patients. Radiographs should be evaluated for abnormal glenoid version, dysplasia, or hypoplasia, and humeral head abnormalities also occasionally can be detected on radiographs. Magnetic resonance imaging provides a clearer picture of the soft tissue anatomy in the shoulder and is often combined with magnetic resonance arthrography to give perhaps the gold standard of anatomic evaluation for MDI. Although labral pathology can certainly be present in patients with MDI, the classic finding is increased capsular volume and a patulous capsule (Fig. 42.4).⁴⁵ Schaeffeler et al.⁴⁶ correlated the clinical diagnosis of atraumatic MDI with redundancy signs on magnetic resonance arthrography with high sensitivity and specificity. The finding of an increase in the rotator interval dimension of patients with MDI has been mixed.^{47,48} A large retrospective review by Lee et al.⁴⁹ demonstrated larger rotator interval and capsular dimensions in MDI shoulders compared to controls. More recent research by Lim et al.⁵⁰ looked at labrocapsular distances on magnetic resonance arthrography and found

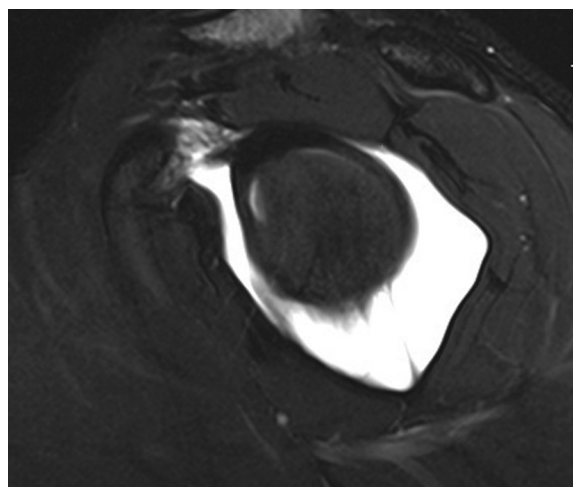


Fig. 42.4 Magnetic resonance imaging scan of a patulous capsule.

the inferior labrocapsular distance to have a high specificity and sensitivity of detecting clinical MDI.

DECISION-MAKING PRINCIPLES

The most important principle of treatment in patients with MDI or any shoulder pathology is an understanding of the activities performed by the patient. Especially in the athlete, a fine line often exists between the advantage of laxity and the limitations of instability. As noted earlier in the history section, understanding how the patient uses the shoulder and the expectations of the patient are essential to guide proper treatment.

In the absence of trauma, MDI usually results when a patient's dynamic stabilizers fail to compensate for a baseline lax joint. Consider a swimmer who presents with a chief complaint of pain after ramping up her training volume in preparation for a league championship, or a baseball pitcher who begins experiencing decreased control in the late innings of a game. In both cases, fatigue of the dynamic stabilizers results in their failure to restrain the baseline laxity, resulting in symptoms. Both scenarios illustrate the point that MDI has a significant dynamic muscular component, and that failure to recognize and treat this component will likely result in an unsatisfied patient. For this reason, conservative rehabilitation remains the mainstay of any initial treatment program of MDI.

Several studies have noted that in young patients with hypermobility, instability symptoms decrease as patients reach skeletal maturity, and thus a prolonged program of physical therapy is strongly recommended.^{51–53} Kronberg et al.⁵⁴ found significant imbalances in muscular control in patients with generalized ligamentous laxity. These investigators noted that muscle-strengthening and scapular rhythm regimens may be more effective than soft tissue reconstructions in these patients. Such a treatment regimen improves tone and proprioceptive control of the deltoid and rotator cuff musculature and should focus on progressive resistance exercises that target the rotator cuff and deltoid muscles.⁵⁵ This targeting results in a substantial increase in rotator cuff activation, which functionally reduces instability.⁵⁶ In addition, scapulothoracic mechanics should be carefully evaluated, and any dyskinesia should be a major focus of the rehabilitation program. The synchronous firing of the serratus anterior, trapezii, rhomboids, and subscapularis position the scapular platform in space and can affect version and inclination, forming a stable base for the functioning of the glenohumeral joint. It is important to stress to the patient that these strengthening programs are a significant commitment, beginning with 6 months to 1 year of therapy, followed by a chronic maintenance program.^{1,7,56}

In a classic study applying these principles, Burkhead and Rockwood⁵⁵ evaluated the results of a rehabilitation program to treat patients with instability. In patients with atraumatic instability, 83% of patients obtained a good or an excellent result with muscle strengthening, and nearly 90% of those with MDI obtained a similar result. However, other reports have been less favorable. Kiss et al.⁵⁷ evaluated a similar group and found good or excellent results in only 61% of their

patients; they noted that patients with a work-related injury or psychological problems were less likely to benefit from the rehabilitation program. Misamore et al.⁵⁸ performed a careful longitudinal study of 64 patients with MDI who underwent physical therapy as their initial treatment. These investigators found that at 2-year follow-up, 66% of patients had either required operative stabilization or reported only fair or poor shoulder outcomes. More discouraging, at a final follow-up of 8 years, only 30% of the cohort had avoided surgical intervention and rated their shoulder as good or excellent, with only eight patients free of pain or instability complaints. In a study by Nyiri et al.⁵⁹ it was found that kinematic parameters and muscle activity patterns can be normalized after surgery and physical therapy, but not with physical therapy alone. Thus although rehabilitation remains a mainstay of treatment for patients with MDI, this condition represents a significant therapeutic challenge and may fail to result in long-term patient satisfaction and return to activity.

TREATMENT OPTIONS

Surgical treatment for MDI should be considered after failure of a long-term rehabilitation program and a frank discussion with the patient about expectations and the need for a postoperative commitment to rehabilitation. The primary pathologic condition in MDI is a loose, redundant capsule, especially inferiorly, and thus the aim of surgical intervention is to reduce the redundancy and effectively shorten the static stabilizers, especially in the primary direction of instability. Techniques, such as capsular shifts, thermal capsulorrhaphy, capsulolabral augmentation, supplementary support, and arthroscopic plication, have been used to accomplish this goal.

Open Inferior Capsular Shift

The open inferior capsular shift was the original operative procedure described by Neer and Foster,¹ in which they performed a humeral-based shift to eliminate capsular redundancy (Fig. 42.5). This procedure could be performed anteriorly or posteriorly or even combined in rare cases to address global capsular redundancy. Patients are positioned in a beach chair-type position, and a standard deltopectoral approach is used. The subscapularis is taken down and the rotator interval between the superior and middle glenohumeral ligaments is closed. A T-shaped humeral-based incision is then made, creating an inferior and superior flap of the anterior capsule, which is elevated from the humeral neck. From here, intra-articular pathology can be identified and addressed. The magnitude of the capsular shift remains an art, balancing the stability gained by reducing the volume of the capsule with expectations of postoperative range of motion (ROM). The inferior capsule is advanced superiorly, and the superior capsule is attached over the top to reinforce this shift. The subscapularis is then reattached separately in its original position, and the procedure is complete after examination reveals appropriate motion without undue stress on the repair. Several investigators have demonstrated that this technique results in high satisfaction rates with low rates of recurrence.^{1,5,6,60–66} However, more modern evaluations of success, including return to sporting activities, have been less enthusiastic.^{63,64,67} Since the

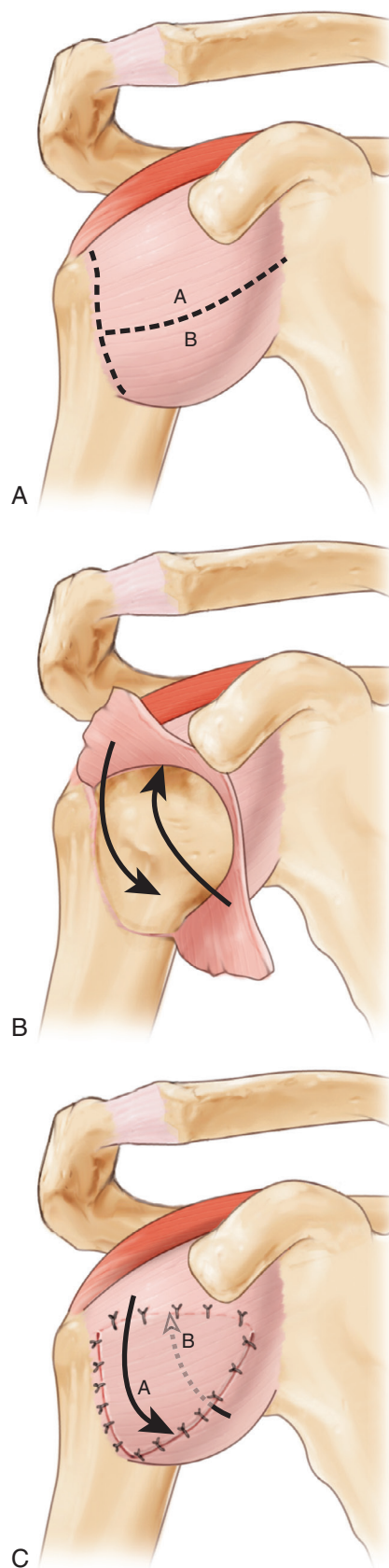


Fig. 42.5 Inferior capsular shift. (A) A T-shaped humeral-based incision is made in the anterior capsule, creating an inferior and superior flap. (B) These flaps are elevated off the humeral neck so they can be shifted. (C) The inferior capsule is advanced superiorly, and the superior capsule is then attached over the top to reinforce the shift.

original description, several modifications to this technique have been reported, including a subscapularis-sparing technique⁵ and a glenoid-based shift.^{67–69}

Arthroscopic Plication

The development of advanced arthroscopic techniques has provided increased visualization of the glenohumeral joint and the ability to approach the pathology on both sides of the glenoid through a single approach, without violating the subscapularis attachment. This technique is especially well suited for patients with posteroinferior MDI, in whom a lack of posterior labral height can decrease resistance to posterior shear stress and contribute to instability in some patients,^{8,70} or for patients with both a posterior and anterior component, for whom traditionally a second approach would be required. Standard posterior and posterolateral portals, along with anterosuperior and midglenoid portals, suffice to address all aspects of instability.

Arthroscopic capsular plication can be performed in either the beach chair or lateral decubitus position. The history and physical examination (previously detailed), as well as an examination while the patient is anesthetized, can help determine the primary direction of instability, which should be addressed first. Wichman and Snyder⁷¹ described an arthroscopic technique for addressing capsular redundancy through plication. After biologic preparation of the capsule with rasping or light débridement, a suture hook loaded with a suture shuttle relay is used to create a pinch stitch inferiorly within the capsule. This stitch is then advanced superiorly and medially through the labrum, creating an overlap that effectively shortens and thickens the capsule. A nonabsorbable suture is then passed in a retrograde fashion, creating the plication stitch. The opposite end can be passed in a similar fashion to create a mattress stitch, if desired. Additional sutures can be placed anteriorly and posteriorly, progressing superiorly until the desired amount of plication is achieved. Several authors have compared the ability of an arthroscopic capsular shift to reduce volume in the shoulder.^{72–74} Ponce et al.⁷³ demonstrated that each plication stitch reduces the volume of the joint by 10% in a linear relationship, and five such stitches equals the volume reduction of a lateral-based open inferior capsular shift. These findings suggest that arthroscopic multiple-pleat plication may be as effective as an open inferior capsular shift for eliminating capsular redundancy.

Sutures can be passed through the capsular tissue and sutured directly to the labrum, or a suture anchor can be used. Provencher et al.⁷⁵ demonstrated that although the strengths of the constructs were similar, plication without suture anchors resulted in more displacement at the labrum and may be preferable. Anchor-based plication was shown to be better biomechanically than suture capsulorrhaphy in a biomechanical study by Kersten et al.⁷⁶

Several authors have investigated these techniques in the clinical setting. Raynor et al.⁷⁷ followed 45 shoulders after arthroscopic pancapsulorrhaphy with suture anchors for a mean of 3.3 years. Kaplan-Meier survivorship at 1 year was 100% and 87% at 3 years. The recurrent instability rate was just under 17%, very similar to prior literature. Comparable results were reported by Alpert et al.⁷⁸ in their review of MDI patients with 270 degrees labral tears. While these two studies only utilized suture anchors, Gartsman et al.⁷⁹ followed up with 47 patients for 35 months

after arthroscopic suture and/or anchor-based plication for MDI. These investigators found good or excellent results in 94% of their patients, with a 2% recurrence rate, and noted that 85% of their patients were able to return to their desired level of sports. Several others have found success with a combination of suture anchors and/or plication sutures with similar results.^{64,74–82}

Thermal capsulorrhaphy was advanced as an alternative to suture methods and provided the advantage of a simple and quick arthroscopic approach to address MDI. The theory was that heating the capsule disrupted the cross-linking of the collagen, allowing it to “crimp” and shorten. Despite early reported success,⁸⁹ other authors have reported high failure rates and the potential for chondrolysis and nerve injury with this technique,^{90–92} and it has largely been abandoned. However, a recent multicenter, randomized, clinical trial by Mohtadi et al.⁹³ compared electrothermal arthroscopic capsulorrhaphy (ETAC) to open inferior capsular shift in patients with MDI. The group found nearly equivalent recurrence rates (1%) and outcomes in The Western Ontario Shoulder Instability Index (WOSI), American Shoulder and Elbow Surgeons (ASES) assessment form, and Constant scores at the 2-year follow-up.

Rotator interval closure remains a staple of the open procedure but is more controversial in the arthroscopic setting. Neer and Foster¹ routinely performed this adjunct at the beginning of their approach, and Harryman and colleagues⁹⁴ elegantly demonstrated that an open medial to lateral imbrication of the coracohumeral ligament greatly reduced inferior and posterior translation. The work by Harryman et al.⁹⁴ is often cited as justification for rotator interval closure in all settings. Other authors have been unable to replicate these findings⁹⁵ using a similar technique. In the arthroscopic setting, a rotator interval closure is most often performed in a superior to inferior fashion. However, cadaveric models have failed to demonstrate that this technique assists in reducing inferior translation,^{96,97} instead, these models have shown that it results in a restriction of external rotation. As rotator interval closure continues to be debated, it should be applied judiciously with an understanding of its limitations.

It should be noted that most of the operative outcomes studies for MDI have short-term follow-up periods and that results may deteriorate over time.¹⁰ Authors from Rowe et al.²⁵ to Krishnan et al.⁹⁸ have recommended a combination of procedures to address the true collagenopathy. Biceps augmentation, allograft reconstruction, and glenoid osteotomy have all been attempted to control stability in patients with collagen abnormalities and MDI, but few studies are available to evaluate clinical outcomes with these alternative approaches.^{10,98} Nonetheless, new techniques are continually being developed to address the problem.⁹⁹

POSTOPERATIVE MANAGEMENT

In general, the risk of recurrence after MDI is higher than the risk of stiffness, and therefore a cautious approach is best. The shoulder is placed in neutral rotation for 6 weeks using an immobilizer with a slight abduction pillow. One of the most critical points of the postoperative program is to understand that immobilization does not mean neglect. Patients are encouraged to begin using the hand, wrist, and elbow immediately after surgery. Further, although shoulder motion is limited for this period, we

Authors' Preferred Technique

Multidirectional Instability

- Understanding each individual patient's condition and expectations is critical in our treatment decision process.
- As previously expressed, making the diagnosis of multidirectional instability versus multidirectional laxity can be difficult, but it is vital for the appropriate treatment.
- In our experience, not all "therapy" is equal, and good communication among the surgeon, patient, and therapist to ensure the proper emphasis on strength and balance can result in excellent outcomes without operative intervention.
- For patients in which conservative management fails, an arthroscopic approach is preferred for its ability to address circumferential pathology and to allow more aggressive reengagement of the shoulder musculature postoperatively.
- We prefer an arthroscopic approach in the lateral position as it offers improved visualization in our experience.

Portals

- Preoperative examination under anesthesia is critical in our assessment of the patient's stability; you must have a baseline.
- All aspects of the glenoid labrum and capsule can be approached through standard posterior, anterosuperior, and midglenoid portals.¹⁰⁰
- We will use a modified posterolateral portal for the proper angle of attack posteriorly and inferiorly as previously described in the literature.¹⁰⁰

Plication Strategy

- After a thorough diagnostic arthroscopy, all identified associated pathology is addressed before the capsular plication. In our experience, it is common to have labral and other pathology in addition to capsular laxity.
- It is critical to biologically prepare the capsule for plication, using either a rasp or a shaver without suction to optimize postoperative healing.
- The plication begins inferiorly and advances superior using each stitch as a reference for the next.
- Capsular redundancy is eliminated with a pinch-tuck technique (Fig. 42.6) while suture anchors are used on the glenoid.
- The anterior and posterior insertions of the inferior glenohumeral cardinal ligament are grasped on either side using double-loaded suture anchors at the 5:30 and 7:30 positions.
- A full-thickness capsular bite is taken with each suture. We are cautious not to get overzealous with the inferior plication stitches as thicker bites increase the risk of axillary nerve injury.
- The amount of plication is approximately 1 cm, but it is dependent on multiple factors, including the examination while the patient is anesthetized, hand dominance, and goal activities.
- As the case proceeds superiorly, either simple or mattress sutures and a combination of suture anchor or plication stitches are utilized, depending on the quality of the glenoid labrum.

Rotator Interval

- We do not routinely perform a rotator interval closure except in revision cases and in patients with known collagen disorders, and when doing so we usually only perform this closure on a patient's nondominant side.
- We have not found plication stitches near the biceps anchor to be helpful and therefore we do not plicate superiorly.

Closure

- Using the preoperative examination under anesthesia as the baseline, the stability exam is repeated upon completion of the plication (Fig. 42.7).
- The portals are closed, dressed sterilely, and a shoulder immobilizer with abduction pillow is applied.

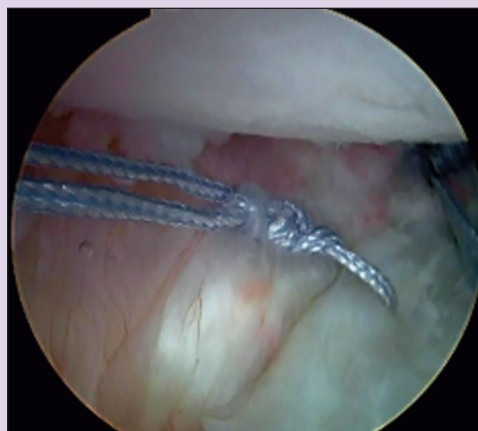


Fig. 42.6 Arthroscopic plication.



Fig. 42.7 Finished plication. At the completion of capsular plication, the glenoid bumper is restored and the capsular redundancy is diminished.

strongly believe that the rotator cuff and periscapular musculature should be dynamically reengaged as soon as possible after surgery. The arthroscopic and open approaches are similar, except that after arthroscopic capsular plication, the subscapularis is reengaged sooner than after an open repair. In our experience, it is rare for stiffness to develop in a patient with MDI, but atrophy of the dynamic stabilizers often persists after surgical intervention without a specific postoperative regimen to simultaneously protect the reconstruction and reactivate these muscle groups. Given the key role that the musculature plays in dynamic stability in persons with MDI specifically, 6 weeks of disuse atrophy must

be avoided. In one study, muscle strength decreased 54% after 7 weeks of immobilization and was only partially regained after remobilization.¹⁰¹ Several authors have demonstrated that postoperative protocols can successfully restore normal muscle activation and motion patterns after capsular shift.^{56,102} Thus early scapular mobilization (Fig. 42.8) is demonstrated to the patient preoperatively, along with isometric strengthening of the rotator cuff and periscapular musculature. Patients are encouraged to begin the exercises on postoperative day 1. In addition, because the shoulder works within a kinetic chain, core strengthening is an essential part of the perioperative strengthening program

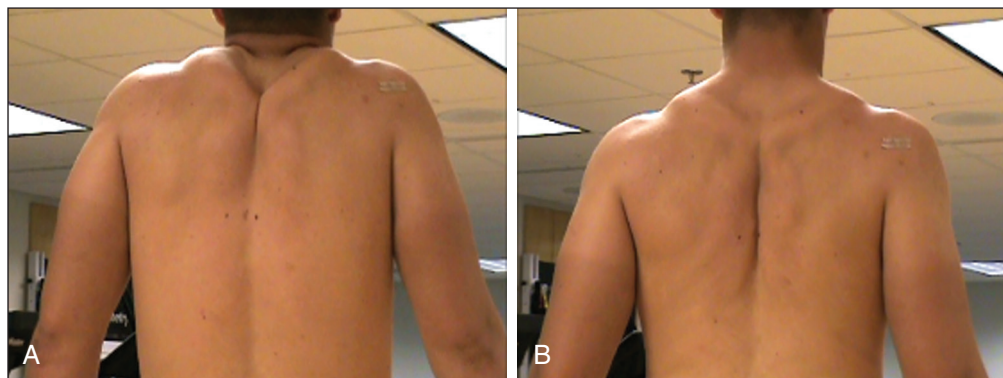


Fig. 42.8 Scapular mobilization. Early mobilization of the scapular musculature is performed to actively reengage the dynamic stabilizers of the shoulder. Here, scapular retraction and elevation are shown in (A) and (B), respectively, and protraction, depression, and rotation are emphasized.

and is actually taught in prehabilitation and is encouraged throughout the postoperative course.

At 6 weeks, the patient is evaluated for stiffness of the shoulder and maintenance of muscle rhythm and strength. At this point, the patient generally is allowed to stop using the sling and progress toward postoperative motion goals. These goals are tailored to the patient's expected needs. The chosen activity, arm dominance, and time frame of the ultimate return to sport are all considerations when making decisions about how quickly to proceed. Muscle strengthening is continued, and if the patient is on track, it is advanced. By 3 months, the patient should have reasonable humeroscapular rhythm with minimal scapular winging or substitution during elevation and abduction. Most of the patient's ROM is generally expected to have returned by this point.

Months 3 to 6 are devoted to fine tuning the desired ROM, and emphasis is placed on functional strength progression. Progression of loads is from low to high; speeds, from slow to fast; and direction, from unidirectional to multidirectional. Finally, the stability of the platform of these exercises can be perturbed to force dynamic co-contraction of the shoulder girdle. The physical therapist or corrective exercise specialist should be given the freedom to vary each of these parameters until the patient is confident and exhibits high performance in each of these aspects. At this point, usually around 6 months after surgery, the patient is cleared to progress to full activity, with emphasis on continued training with endurance activities so that fatigue won't precipitate decompensation of the dynamic stabilizers and compromise the result.

In the case of athletes, return-to-play (RTP) decisions should be made as a team. The surgeon must be confident that the reconstruction is healed and that the patient has achieved the necessary dynamic protection of the shoulder that was surgically repaired. The therapist or corrective exercise specialist is perhaps the most important decision maker for return to sport. This team member spends the most time with the athlete and learns to recognize how the athlete compensates or demonstrates poor dynamic control of the shoulder. It is the job of the therapist not only to help the patient regain strength in the shoulder girdle but also to ensure that this activity is placed well within an

effective kinetic chain so that return to sport will not result in reinjury. The athlete also has an important role in the RTP decision. A patient's confidence in the stability of his or her shoulder, as well as open communication of any signs of trouble, are critical elements of the decision to return to activity that the patient alone can provide.

RESULTS

The results of the operative treatment of MDI are summarized in Table 42.3. Neer and Foster¹ originally reported a 97% satisfactory rate with the open inferior capsular shift, and this finding has been repeatedly borne out in the literature. However, patients with voluntary instability and an underlying psychiatric diagnosis are poor candidates for any operative approach.^{1,10,25} It is also important to note that most of the reported series are relatively short term, and recurrence rates may increase with longer term follow-up.¹⁰ Finally, the definition of MDI continues to be inconsistently applied. For example, the series by Altchek and colleagues⁶⁷ did not include any patients with generalized ligamentous laxity, yet it was described as a population of patients with MDI. The series by Kim and colleagues⁷⁰ consisted of patients with posteroinferior instability, and it was noted that all patients had a labral lesion, which is inconsistent with the definition provided by others.³¹ A traumatic etiology has been frequently reported,⁶⁰ which is not a classic presentation in persons with MDI as defined by other authors.² Thus comparing findings across series is difficult, and surgeons should be careful not to oversimplify the diagnosis of MDI.

The arthroscopic approach to MDI has shown similar success rates to that of the classic open inferior capsular shift. In a systematic review of the MDI literature, Longo et al.¹⁰³ concluded that arthroscopic capsular plication and open capsular shift were the best procedures for MDI treatment after failed conservative management, and the results between the two modalities were comparable. The largest known meta-analysis of the literature by Chen et al.¹⁰⁴ found no difference between recurrent instability rates between open capsular shifts and arthroscopic capsular plications. However, Chen and colleagues did find a difference in postoperative stiffness between the two groups.

TABLE 42.3 Results for Operative Treatment of Multidirectional Instability

	Author	Follow-Up	Technique Details	Outcome Measure	Success Rate	Complications/Notes
Open inferior capsular shift	Neer and Foster ¹	<2 years	Open inferior capsular shift	Recurrence	98% satisfactory	None
	Pollock et al. ⁶³	5.1 years	Open inferior capsular shift	Satisfaction/recurrence	94% good/excellent; 4% recurrence	
	Cooper and Brems ⁶	3.3 years	Open inferior capsular shift	Satisfaction/recurrence	86% improved; 10% recurrence	15% believed their shoulder deteriorated with longer term follow-up
	Lebar and Alexander ⁶²	2.3 years	Open inferior capsular shift	Satisfaction/recurrence	All improved; 10% recurrence	Military population
	Hamada et al. ⁶¹	8.3 years	Open capsular shift	Rowe/recurrence	59% good/excellent; 26% recurrence	Mixed population of MDI and anteroinferior instability
	Choi and Ogilvie-Harris ⁶⁰	3.5 years	Open inferior capsular shift	ASES/recurrence	92% success for anterior repairs (8% recurrence); 81% success for posterior repairs (12% recurrence)	>75% of contact athletes returned to sport except those with bilateral repairs, of whom 17% returned to sport
	Bak et al. ⁵	4.5 years	Open inferior capsular shift	UCLA/Rowe/recurrence	88% good/excellent; 8% recurrence	Return to sport rate of 84%, including 76% in overhead throwing athletes
	Vavken et al. ⁶⁴	7.5 years	Open inferior capsular shift	ASES/QuickDASH/recurrence	87% satisfied/very satisfied; 47% recurrence	Five Ehlers Danlos patients; one perioperative reflex sympathetic dystrophy; rotator interval closure did not affect stability, satisfaction, return to sport, or recurrence
	Bigliani et al. ⁶⁵	4 years	Open inferior capsular shift	Patient reported based on pain, stability, range of motion, and return to sport/recurrence	94% good/excellent; 3% recurrence	Anterior-inferior instability; one transient musculocutaneous neuropraxia
	van Tankeren et al. ⁶⁶	3.3 years	Open inferior capsular shift	Rowe/Constant/12 item Dawson questionnaire/satisfaction/recurrence	82% excellent Rowe scores; 88% satisfied; 24% recurrence	53% concomitant labral pathology
	Mohtadi et al. ⁹³	2 years	ETAC vs open inferior capsular shift	WOSI/ASES/Constant/range of motion/recurrence	7% recurrence for ETAC and 14% for open inferior capsular shift (no statistical difference)	Included MDI and MDL with primary anterior inferior instability; three post-op adhesive capsulitis/stiff shoulders; (one ETAC, two open inferior capsular shift)
	Altchek et al. ⁶⁷	3 years	T-plasty modification of inferior capsular shift	Satisfaction/recurrence	95% satisfied; 10% recurrence	Only half had general ligament laxity; throwers did not return to sport reliably
Arthroscopic capsular plication	Marquardt et al. ^{68a}	7.4 years	T-plasty modification of inferior capsular shift with suture anchors	Rowe/recurrence	89% good/excellent; 10% recurrence	Anterior-inferior instability
	Lyons et al. ⁶⁹	2.3 years	Laser-assisted arthroscopic plication	Neer criteria/stability	96% stable and asymptomatic	Added plication stitches to interval in many patients
	Duncan and Savoie ⁸³	1–3 years	Arthroscopic inferior capsular shift	Neer criteria/stability	All patients satisfactory, 90 on Bankart score	10 patients

TABLE 42.3 Results for Operative Treatment of Multidirectional Instability—cont'd

	Author	Follow-Up	Technique Details	Outcome Measure	Success Rate	Complications/Notes
Arthroscopic capsular plication	Gartsman et al. ⁷⁹	2.9 years	Arthroscopic plication	ASES/Constant/Rowe/UCLA	94% good/excellent; 2% recurrence ASES/Constant/UCLA all improved significantly	
	Baker et al. ⁸²	2.8 years	Arthroscopic capsular plication	ASES/WOSI	ASES: 94; WOSI: 91% normal	
	Treacy et al. ⁸⁵	5 years	Arthroscopic plication Caspari technique	Rowe/recurrence	88% satisfactory	Two patients with persistent pain from posterior knots
	McIntyre et al. ⁸⁴	2.8 years	Arthroscopic multiple suture technique	Rowe/recurrence	94% good/excellent; 5% recurrence	One transient musculoskeletal sensory neuropathy
	Kim et al. ⁷⁰	4.3 years	Arthroscopic suture plication	Rowe/recurrence	94% good/excellent results; 3% recurrence	Patients had posteroinferior MDI, all with a labral lesion
	Witney-Lagen et al. ⁸⁶	5.1 years	Arthroscopic suture plication with purse stringing	OIS/recurrence	95% good/excellent; 4% recurrence	Rotator interval closure included in 10% of patients; one postop stiffness; one superficial wound infection
	Ma et al. ⁸⁷	3 years	Arthroscopic pancapsular plication with sutures	ASES/Constant/Rowe/VAS/recurrence	100% good/excellent; no recurrence	All patients had bilateral laxity; rotator interval closures performed in all patients
	Raynor et al. ⁷⁷	3.3 years	Pancapsular capsulorrhaphy with labral repair using suture anchors	ASES/QuickDASH/SF-12 PCS/SANE/recurrence	Kaplan–Meier survivorship 100% at 1 year and 87% at 3 years; 17% recurrence	29% of shoulders had rotator interval closures; SLAP and Bankart repairs were also included
	Buess et al. ⁸⁰	2.3 years	Arthroscopic repair with anchors and/or capsular plication	Constant/SST/Rowe Score Westpoint-Modification/VAS/recurrence	Significant improvement in Rowe score in all patients; 14% recurrence	Four instability groups with seven patients classified as predominantly posterior multidirectional instability
	Voigt et al. ⁸¹	3.3 years	Arthroscopic anteroposterio-inferior capsular plication with sutures/anchors and interval closure	SST/Constant/Rowe/VAS/recurrence	77% good/excellent Rowe scores at final follow up, 33% required reoperation	Eight patients; all patients were MDI with hyperlaxity (Gerber B5)
	Alpert et al. ⁷⁸	4.7 years	Arthroscopic capsular plication with suture anchors	WOSI/ASES/SST/VAS/satisfaction/recurrence	85% mostly/completely satisfied; 15% recurrence	All patients had a minimum of 270 degrees arthroscopic labral repair; Rotator interval closed in 38% of patients
	Hewitt et al. ⁸⁸	4.8 years	Arthroscopic pancapsular plication with sutures	ASES/Constant/Rowe/UCLA/WOSI/satisfaction/recurrence	97% mostly/extremely satisfied; 83% good/excellent Rowe scores	Three post-op traumatic dislocations

^aIncludes data from Steinbeck J, Jerosch J. Surgery for atraumatic anterior-inferior shoulder instability. A modified capsular shift evaluated in 20 patients followed for 3 years. *Acta Orthop Scand.* 1997;68(5):447–450.

ASES, American Shoulder and Elbow Surgeons; DASH, disabilities of the arm, shoulder, hand; ETAC, electrothermal arthroscopic capsulorrhaphy; MDI, multidirectional instability; MDL, multidirectional laxity; OIS, Oxford Instability Score; SANE, single assessment numeric evaluation; SF-12 PCS, short form-12 physical component summary; SLAP, superior labrum anterior to posterior; SST, simple shoulder test; UCLA, University of California–Los Angeles; VAS, visual analog scale; WOSI, Western Ontario Shoulder Instability.

On average, patients lost 5 degrees more of external rotation after an open capsular shift compared to an arthroscopic capsular plication. They also recommended against thermal capsular shrinkage in the treatment of MDI secondary to high failure rates.¹⁰⁴

Return to Sport

Several studies have reported return to sports as a specific outcome measure after the surgical treatment of MDI (Table 42.4). The reported success rate varies from 17% to 97% and reflects variation in inclusion criteria and specific sport requirements.

TABLE 42.4 Return to Sport After Operative Treatment of Multidirectional Instability

Author	Technique	Outcome Measure	Success Rate
Bak et al. ⁵	Open inferior capsular shift	Return to previous level of sport	Return to sport 84%, including 76% in overhead throwing athletes
Choi and Ogilvie-Harris ⁶⁰	Open inferior capsular shift	Return to previous level of sport	Anterior: 82%; posterior: 75%; bilateral: 17%
Vavken et al. ⁶⁴	Open inferior capsular shift	Return to previous level of sport	69% returned to sport; 38% returned to previous level of sport
Bigliani et al. ⁶⁵	Open inferior capsular shift	Return to previous level of sport	92% returned to sport; 75% returned to previous level of sport
Altchek et al. ⁶⁷	T-plasty modification of inferior capsular shift	Return to sport	83% returned to sport, but all throwers had decreased velocity
Marquardt et al. ⁶⁸	T-plasty modification of inferior capsular shift with suture anchors	Return to previous level of sport	72% including ⅓ elite athletes
Lyons et al. ⁸⁹	Laser-assisted capsular plication	Return to previous level of sport	86%
Gartsman et al. ⁷⁹	Arthroscopic inferior capsular shift	Return to previous level of sport	85%
Baker et al. ⁸²	Arthroscopic inferior capsular shift	Return to sport with little or no limitation	86%
McIntyre et al. ⁸⁴	Arthroscopic inferior capsular shift	Return to previous level of sport	95% return to previous level, but no elite throwers
Witney-Lagen et al. ⁸⁶	Arthroscopic suture plication with purse stringing	Return to previous level of sport	90% return to previous level, but no professional athletes
Hewitt et al. ⁸⁸	Arthroscopic pancapsular plication with sutures	Return to sport	97% returned to sport with 20% at a reduced level
Ma et al. ⁸⁷	Arthroscopic pancapsular plication	Return to previous level of sport	22%; study included professional/amateur overhead athletes
Raynor et al. ⁷⁷	Pancapsular capsulorrhaphy with labral repair using suture anchors	Return to previous level of sport	77% returned to or slightly below their preoperative level of sport
Voigt et al. ⁸¹	Arthroscopic anteroposterio-inferior capsular plication with interval closure	Return to previous level of sport	78% returned to sport; 44% returned to previous level of sport

McIntyre et al.⁸⁴ reported a 95% return to sport after arthroscopic treatment for MDI, but these authors were careful to point out that their series did not contain elite throwers. A Ma et al.⁸⁷ study on professional and amateur overhead athletes noted a 100% RTP, but only 22% returned to their previous preoperative level. Bak et al.⁵ also reported on overhead throwing athletes and found that only 76% resumed throwing. Direction and approach also seem to affect return to athletics. Choi and Ogilvie-Harris⁶⁰ reported an 82% return to sport for patients with an anterior approach, but contrasted this finding with a 17% return with a bilateral approach. It is likely that patients with a true collagen disorder may self-limit their activities long before presentation for surgical consideration, and thus the favorable results reported in the literature should be carefully applied to patients who had previously been able to achieve active sport participation.

Complications

Although the reported complication rates after the operative treatment of MDI remain low, several types of complications have been reported—the most feared of which is neurologic injury, although reported rates are uncommon and generally self-limiting throughout the literature. A thorough understanding of the proximity of neurologic structures is essential to minimize the risk of this postoperative complication. McFarland et al.¹⁰⁵ reported on the proximity of the applicable neurovascular

structures for instability surgery. These investigators found the plexus to be as close as 5 mm from the glenoid rim. They also noted that retractors placed superficial to the subscapularis muscle or used along the scapular neck make contact with the brachial plexus in all positions. Other authors^{106,107} have noted that the axillary nerve position averages 3.2 mm from the capsule of the inferior glenoid. Applicable to the posterior approach, the suprascapular nerve has been shown to be an average of 18 mm from the midportion of the posterior glenoid rim.¹⁰⁸

Clinically, a total of seven axillary nerve injuries have been reported after treatment for MDI. Neer and Foster¹ reported three transient axillary neurapraxias in their original series of open inferior capsular shifts, and Miniaci and McBurnie⁹² reported on four patients with transient neurapraxia after arthroscopic thermal capsulorrhaphy, including one patient with deltoid weakness. In both series, symptoms resolved for all patients without further intervention. Several authors have also reported transient neurapraxia of the musculocutaneous nerve, without persistence of symptoms,^{60,65,67,79} and one median nerve neurapraxia has been reported.⁶³ Vavken et al.⁶⁴ had one case of reflex sympathetic dystrophy in their series of patients undergoing open inferior capsular shift. A meticulous anatomic understanding, along with the judicious use of retractors, is perhaps the best defense against these complications.¹⁰⁵

Although postoperative stiffness is generally not a significant issue in patients with MDI, overaggressive plication can result

in motion loss, especially in external rotation,^{95,97,109} and several cases of adhesive capsulitis has been reported after arthroscopic treatment with thermal capsulorrhaphy for MDI.^{92,93,110}

Recurrent instability is a far more common negative outcome than postoperative stiffness after treatment for MDI. When surgical intervention fails, it is generally accompanied by persistence or recurrence of capsular laxity. Zabinski et al.¹¹¹ evaluated 20 patients with MDI who underwent revision stabilization. At the time of the revision, the primary lesion was capsular laxity in all patients. At the 5-year follow-up, only 45% of patients had a good or excellent result, despite multiple reoperations.¹¹¹ Levine et al.¹¹² also reported on a series of revision surgeries for failed anterior stabilizations. Twenty-two percent had an unsatisfactory result, and atraumatic causes of failure, voluntary dislocations, and unaddressed capsular laxity—all signs of MDI—were noted risk factors associated with poorer results. A case series from Dewing et al.¹¹³ analyzed revision patient outcomes after anterior capsulolabral reconstruction with tibialis anterior or hamstring allograft for recurrent shoulder instability. Although the patient population was not homogeneous, it included 13 patients with the diagnosis of MDI. At 3.8 years mean follow-up, 45% of shoulders remained completely stable. Some revision surgery series have been more successful when thermal capsulorrhaphy was the primary surgery. Massoud et al.¹¹⁴ treated a small series of these patients with an open inferior capsular shift procedure and demonstrated improved outcome scores in all their MDI patients.

FUTURE CONSIDERATIONS

The treatment of MDI remains a challenging proposition. Although the reported success rates in the literature are consistently favorable for the operative treatment of MDI, expert opinion from experienced surgeons often reflects more cautious optimism. The lack of a comprehensive classification system hinders the ability to compare data across studies and would be an excellent future direction. Such a classification has been suggested by several authors,^{3,31,115} but none has gained universal traction. Another future direction that would be of great benefit is a more complete evaluation of nonoperative management for MDI. All authors recommend a trial of physical therapy, but the length of program, specific recommendations, and results are mixed. Comparative studies to more clearly define how long a nonoperative trial should last and the risk factors for failure would go a long way in assisting the clinician in management decisions.

As previously mentioned, literature reviews have shown similar outcomes between arthroscopic and open treatment methods for MDI. However, there still is no direct comparison of arthroscopic versus open management. Furthermore, long-term clinical studies regarding suture material and anchors are lacking, specifically, knotless versus knotted plication sutures. Although studies have included return to sport as an outcome measure, further research is needed to clarify inclusion criteria and specific sport requirements. A long-term comprehensive return to sport and/or activity study would be very beneficial in the treatment of high-level athletes as well as the “weekend warriors.” Defining the amount of stability required while

maintaining an acceptable ROM should also be an area of future research.

Finally, very few studies address the patient with recurrent or persistent MDI. MDI gets much of its infamous reputation from surgeons who have been challenged by patients with MDI for whom multiple surgical attempts at stabilization have failed, and it is somewhat surprising to contrast the near-uniform good and excellent results reported in the literature given the number of failed surgeries experienced by many patients with MDI. Risk factors for failure and definitive reconstructive treatments in the revision setting are an excellent future goal of research. Although allograft reconstructions have been described in the revision setting, further research on this topic may provide surgeons with another tool in their armamentarium for treating MDI.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Neer CS II, Foster CR. Inferior capsular shift for involuntary inferior and multidirectional instability of the shoulder. A preliminary report. *J Bone Joint Surg Am*. 1980;62(6):897–908.

Level of Evidence:

IV, retrospective case series

Summary:

This article reviews the initial results of 36 patients treated for multidirectional instability of the shoulder with an open inferior capsular shift, which was designed to decrease the overall volume of the shoulder joint. Only one of the patients had recurrent subluxation. This article defined the gold standard for treatment of this condition, and it remains the gold standard today.

Citation:

Burkhead WZ Jr, Rockwood CA Jr. Treatment of instability of the shoulder with an exercise program. *J Bone Joint Surg Am*. 1992;74A(6):890–896.

Level of Evidence:

IV, retrospective case review

Summary:

This article investigated results for 140 shoulders with instability that were treated with a specific exercise program. The authors found that only 16% of patients with a traumatic etiology had a good or excellent result, whereas 80% of patients with an atraumatic etiology had a good or excellent result. The authors concluded that every effort should be made to define the etiology of the instability, and if the etiology is atraumatic, the case should be managed conservatively.

Citation:

Gartsman GM, Roddey TS, Hammerman SM. Arthroscopic treatment of multidirectional glenohumeral instability: 2- to 5-year follow-up. *Arthroscopy*. 2001;17(3):236–243.

Level of Evidence:

IV, prospective case series without control

Summary:

This article reports the first arthroscopic treatment of multidirectional instability (MDI). The authors evaluated 47

patients with MDI, reporting 94% good and excellent results at 35 months follow-up. Functional scores improved: the constant from 60 to 92, the Rowe score from 14 to 92, and the University of California–Los Angeles score from 17 to 33.

Citation:

Harryman DT II, Sidles JA, Harris SL, et al. The role of the rotator interval capsule in passive motion and stability of the shoulder. *J Bone Joint Surg Am.* 1992;74A(1):53–56.

Level of Evidence:

Controlled laboratory study

Summary:

This study evaluated the effect of alteration of the rotator interval on translation of the shoulder. The authors found that the shoulder capsule in the region of the rotator interval played a significant role in the range of motion and stability of the shoulder. Release of the interval improved flexion and external rotation, and imbrication resulted in decreased posterior and inferior translation.

Citation:

Misamore GW, Sallay PI, Didelot W. A longitudinal study of patients with multidirectional instability of the shoulder with seven- to ten-year follow-up. *J Shoulder Elbow Surg.* 2005;14(5):466–470.

Level of Evidence:

IV, retrospective consecutive case series

Summary:

This article reports on 64 consecutive patients diagnosed with multidirectional instability (MDI) of the shoulder who were followed up at both 2 and 8 years after completing a conservative physical therapy program. The authors reported that treatment was considered to have failed for 19 of 36 patients; only 14% of patients were free of pain and instability at final follow-up; and the results of this treatment deteriorated between the 2- and 7-year follow-up. The authors questioned the effectiveness of a nonoperative management program to successfully treat MDI.

Citation:

Warner JJ, Deng XH, Warren RF, et al. Static capsuloligamentous restraints to superior-inferior translation of the glenohumeral joint. *Am J Sports Med.* 1992;20(6):675–685.

Level of Evidence:

Controlled laboratory study

Summary:

The authors performed a cadaveric study evaluating the superior–inferior translation of the humerus after selected capsuloligamentous releases. They noted that the primary restraint to inferior translation is the superior glenohumeral ligament and that progressive abduction of the glenohumeral joint results in transfer of the main static restraints to lower in the capsule. The anterior glenohumeral ligament is most critical at 45 degrees of abduction, whereas the posterior portion was more dominant at 90 degrees of abduction in neutral rotation.

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